

BIOLOGY BOARD QUESTION PAPER: JULY 2023**Time: 3 Hrs. Max. Marks: 70****General Instructions:**

The question paper is divided into four sections.

(1) Section A: Q.1 contains Ten MCQs of one mark each.

(i) For MCQs, it is mandatory to write the correct answer with its alphabet (A/B/C/D).

(ii) In MCQs, evaluation will be done for the first attempt only.

Q.2 contains Eight very short answer type questions of one mark each.

(2) Section B: Q.3 to Q.14 are short answer type questions carrying two marks each.

(Attempt any Eight)

(3) Section C: Q.15 to Q.26 are short answer type questions carrying three marks each.

(Attempt any Eight)

(4) Section D: Q.27 to Q.31 are long answer type questions carrying four marks each.

(Attempt any Three)

(5) Begin the answer of each section on a new page.

SECTION – A**Q.1 Select the correct alternatives and write the answers: [10]**

i. If two populations differ in structure of reproductive organs, the isolation is ____.

(A) ethological (B) seasonal (C) mechanical (D) habitat

ii. The primary precursor of Indole-3-Acetic acid is ____.

(A) Tryptophan (B) Phenyl alanine (C) Mevalonic acid (D) Methionine

iii. If one DNA molecule undergoes PCR (3 min/cycle), after 5 cycles, total molecules = ?

(A) 10 (B) 15 (C) 32 (D) 64

iv. The specific gravity of CSF is ____.

(A) 1.005 (B) 1.02 (C) 1.502 (D) 1.81

v. Cardiac output = 5400 ml/min, HR = 72/min. Stroke volume = ?

(A) 65 ml (B) 74 ml (C) 75 ml (D) 78 ml

vi. Detritus food chain starts from ____.

(A) producers (B) dead organic matter (C) parasite (D) photosynthesis



vii. Plants absorb _____ water.

(A) gravitational (B) capillary (C) combined (D) hygroscopic

viii. Organisms tolerant to wide salinity range are _____.

(A) stenothermal (B) euryhaline (C) stenohaline (D) eurythermal

ix. Hisardale breed is developed by crossing _____.

(A) Bikaneri ram × Marino ewe (B) Bikaneri ram × Bikaneri ewe
(C) Marino ram × Bikaneri ewe (D) Marino ewe × Marino ram

x. Perforins are secreted by _____.

(A) Helper T-cells (B) Cytotoxic T-cells (C) Suppressor T-cells (D) Memory T-cells

Q.2 Answer the following questions: [8]

i. Mention the chromosome number having mutated gene for β -Thalassemia.

ii. Which organ produces calcitriol?

iii. If MMC has 26 chromosomes, total chromosomes in endosperm = ?

iv. Define – Facilitated diffusion.

v. Give reason – Energy pyramid is always upright.

vi. What is the base sequence on the DNA template strand coding for methionine?

vii. Deficiency of which element causes Brown heart disease?

viii. Where are Rauber's cells situated in human blastocyst?

SECTION – B

(Attempt any EIGHT) [16]

Q.3 i. Give one example each of:

a. Autosomal dominant traits

b. Autosomal recessive traits

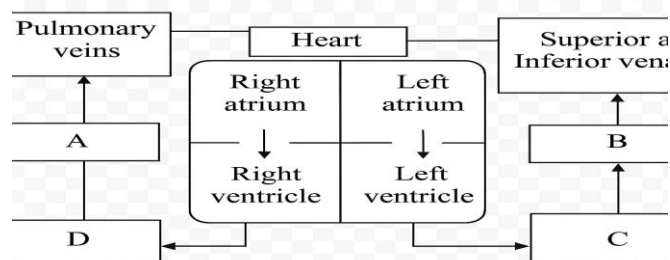
ii. Carrier woman × colourblind man → Show phenotypes in chart.

Q.4 Draw diagrams showing:

i. deletion

ii. inversion

Q.5 Observe double circulation diagram and identify A, B, C, D.



- Q.6 i. Select freshwater fishes: Sardinella, Rastrelliger, Cirrhina, Harpadon, Labeo
ii. Write any two economic importance of Lac.
- Q.7 i. Define – Ecological succession.
ii. Give reason for eutrophication.
- Q.8 i. Mention any two ill-effects of UV rays on humans.
ii. Give significance of Ecosan.
- Q.9 What are oral vaccines? Enlist the benefits.
- Q.10 What is vernalization? Give advantages.
- Q.11 Enlist causes of biodiversity losses.
- Q.12 Give role of hormones Relaxin and Inhibin.
- Q.13 Sketch and label diagram of stoma with kidney-shaped guard cells.
- Q.14 Explain southern blotting and hybridization in DNA fingerprinting.

SECTION – C

(Attempt any EIGHT) [24]

- Q.15 i. Define – Palaeontology.
ii. Give any four significances.
- Q.16 i. What is pollination?
ii. Differentiate between Anemophily and Entomophily w.r.t:
a. pollinating agent b. stigma c. nectar d. fragrance
- Q.17 i. Differentiate between hypotonic and hypertonic solutions.
ii. Effect of exo-osmosis and endo-osmosis on cell shape.
iii. Difference between symplast and apoplast pathway.
- Q.18 i. Control measures to prevent ascariasis.
ii. Complete and rewrite table:

Sr. No.	Name of disease	Name of pathogen
a.	_____	Entamoeba histolytica
b.	Typhoid	_____
c.	_____	Wuchereria bancrofti
d.	_____	Plasmodium species

- Q.19 i. Define – Adaptation.
ii. Explain two adaptations each in Opuntia and Seal.



Q.20 Match and rewrite:

Column-I	Column-II
i. Himgiri	a. Chilli
ii. Pusa shubhra	b. Wheat
iii. Pusa sadabahar	c. Cauliflower
	d. Cotton

Q.21 i. Explain qualitative and quantitative aspects of growth.

ii. Explain phase of cell maturation.

Q.22 i. What is co-dominance?

ii. Red cow (female) $\times$ White bull (male) $\rightarrow$ Show F₂ generation with genotypes.

Q.23 i. Give any two involuntary vital functions of medulla oblongata.

ii. Mention two functions of spinal cord.

Q.24 i. Define – Transcription.

ii. Write anticodons for: AUG, GAG, CUA, CCU

Q.25 i. Position of: a. Eustachian valve b. Bicuspid valve

ii. Differentiate open and closed circulation w.r.t:

a. blood pressure b. exchange of material

Q.26 Draw neat diagram of Graafian follicle. Label oocyte and antrum. Explain structure.

SECTION – D

(Attempt any THREE) [12]

Q.27 i. What is placenta?

ii. Why placenta is considered a temporary endocrine gland?

iii. Significance of hCG.

Q.28 Describe structural and hormonal changes during ovarian cycle.

Q.29 i. Name formed elements useful in blood coagulation. Give normal count per mm³.

ii. Comment on shape and secretion.

iii. Explain mechanism of blood coagulation.

Q.30 i. Full form of BAC and YAC.

ii. Write palindrome for Eco RI and indicate recognition sequence with arrow.

iii. Give any four uses of gene therapy.

Q.31 Explain any four contrivances to prevent self-pollination with examples.

